

Lab 1 — Use-Case Flow Specification

Problem Statement #34 — Digital Gift Card & Loyalty Point Exchange

Use Case	UC-03 Redeem Digital Gift Card	Primary Actor	Account Holder
Includes	UC-06 Generate Voucher Code	Extended by	UC-07 Lock Voucher (Anti-Fraud Check)

Preconditions

- Account Holder is authenticated and logged into the digital wallet platform.
- Account Holder has a non-negative universal exchange credit balance in their wallet.
- The digital gift card catalog (participating Merchant Partners) is available and up to date.

Postconditions

Success: The exchange-credit balance is debited by the gift card's credit cost, a single-use voucher is generated and locked as "redeemed," a digital gift card is issued to the Account Holder, and a confirmation record is logged.

Failure: No credits are debited, no voucher is marked redeemed, and the Account Holder is notified of the reason for failure.

Main Success Scenario

1. Account Holder selects "Redeem Gift Card" from the wallet dashboard.
2. System displays the available digital gift card catalog with each card's exchange-credit cost.
3. Account Holder selects a gift card and confirms the quantity.
4. System checks the Account Holder's exchange-credit balance against the selected card's cost.
5. System generates a single-use voucher code for the redemption (<> UC-06 Generate Voucher Code).
6. System performs an anti-fraud lock on the voucher and debits the exchange-credit cost from the Account Holder's balance.
7. System issues the digital gift card, bound to the locked voucher code, to the Account Holder.
8. System displays an on-screen confirmation and emails the digital gift card and voucher code to the Account Holder.
9. Use case ends successfully.

Alternate Flow A — Insufficient Credit Balance (branches at step 4)

- 4a1. System displays an "Insufficient balance" message and the shortfall amount.
- 4a2. System suggests converting additional loyalty points to exchange credits.
- 4a3. Account Holder either navigates to UC-02 Convert Points to Exchange Credits or cancels the redemption.
- 4a4. Use case ends without redemption; no credits are debited.

Alternate Flow B — Voucher Lock / Anti-Fraud Check Fails (branches at step 6)

- 6a1. Anti-fraud check flags the voucher as a duplicate, malformed, or suspicious (e.g., rapid repeat redemption attempts).
- 6a2. System cancels the transaction and reverses any provisional credit deduction.
- 6a3. System notifies the Account Holder that redemption failed and invites a retry after a cool-down period.
- 6a4. Use case ends in failure; the flagged voucher is retained in the audit log for investigation.